Probability and Statistics B - Practical Sheet 2

Getting started:

1. Open **RStudio** (do not use the base R application).
2. From the menu, select **File** → **New File** → **R Script**. This will create a blank script where you can type and run your R commands.
3. Immediately save your script with a meaningful name, e.g., `Practical1.R`, using **File** → **Save As**. Choose a location you can easily find later.
4. Type your commands in the script and run them using:
   - **Ctrl + Enter** (Windows) or
   - **Cmd + Enter** (Mac)
5. When you have your answers, copy the relevant output (not the entire script) into a Word document.

Important: Always keep your R script saved. This is good practice and ensures you have a record of your work.

1. In R, generate a population of 10000 from a normal distribution with mean 50 and standard deviation 6 using:

   ```
   xpop = rnorm(10000, 50, 6)
   ```

   You wish to create 100 samples from this population each of size 4. One way of doing this is using the following code:

   ```
   x1 = sample(xpop, 100, replace = FALSE)
   x2 = sample(xpop, 100, replace = FALSE)
   x3 = sample(xpop, 100, replace = FALSE)
   x4 = sample(xpop, 100, replace = FALSE)
   m1 = cbind(x1, x2, x3, x4)
   ```

   You can view your first two samples by typing:

   ```
   m1[1,]
   m1[2,]
   ```

   You can also view the means of the first two samples by typing:

   ```
   mean(m1[1,])
   mean(m1[2,])
   ```

   The following steps will allow you to check the **Central Limit Theorem (CLT)** for the distribution of sample means from a normal distribution.

   We need a vector of the averages of each sample. This can be found using:

   ```
   xbar = apply(m1, 1, mean)
   ```

   Obtain histograms (on the same scale) of `xpop` and `xbar` and appropriate numerical statistics to justify the Central Limit Theorem (CLT) in the case of samples of size 4. Also consider normal Q-Q plots and boxplots, commenting on what they show. (Hint: If Q-Q plot is a straight line the data are likely to be from a normal distribution).

```
qqnorm(xbar)
boxplot(xpop, xbar)
```

Hints:

- You can partition the graphics window into 1 row and 2 columns by typing:

  ```
  par(mfrow = c(1, 2))
  ```

- Reset to the usual format with:

  ```
  par(mfrow = c(1, 1))
  ```

- You can force your histogram to cover the range [10, 90] using:

  ```
  hist(xbar, xlim = c(10, 90))
  ```

(a) What do you think is the distribution of `xbar`?

(b) How do the means and standard deviations of `xbar` and `xpop` compare?

(c) Repeat the exercise for samples of size 9 from a normal distribution with mean 500 and standard deviation 100.

2. Suppose that a random sample $X_1, X_2, \ldots, X_6$ is to be collected from a population following a Gamma $(2, 1)$ distribution. We want to investigate the properties of the sampling distribution of two possible estimators of the mean of the population, given by

$$\hat{\theta}_1 = \bar{X}, \quad \hat{\theta}_2 = \frac{9}{30}(X_1 + X_2 + X_3) + \frac{1}{30}(X_4 + X_5 + X_6).$$

(a) Simulate 10000 samples of size $n = 6$ from this population (i.e. from a Gamma $(2, 1)$ distribution). For each sample calculate (and store) the values of the estimators $\hat{\theta}_1$ and $\hat{\theta}_2$. One way to do this in R is by using a "for" loop, e.g using:

```
nsim = 10^4
# Initialise vectors as arrays of 0's
theta.1 = rep(0, nsim)
theta.2 = rep(0, nsim)

for (i in 1:nsim) {
  x = rgamma(n = 6, shape = 2, rate = 1)
  theta.1[i] = mean(x)
  theta.2[i] = (9/30) * sum(x[1:3]) + (1/30) * sum(x[4:6])
}
```

(b) Recall that you have shown in Question 3 of Tutorial 1 that if $X_1, X_2, ..., X_n$ are independent and identically distributed random variables, each with a Gamma (t, λ) distribution, the sample mean has the distribution $\bar{X} \sim$ Gamma $(nt, n\lambda)$; so in this case, $\bar{X} \sim$ Gamma $(12, 6)$. Plot a histogram of the sampling distribution of $\hat{\theta}_1 = \bar{X}$. Does your plot of the sampling distribution of $\hat{\theta}_1 = \bar{X}$ verify this rule? Try to add the pdf of the Gamma $(12, 6)$ distribution on your histogram to see this more clearly. (The code below will help you).

```
hist(theta.1, xlim = c(0, 5), ylim = c(0, 0.7), freq = FALSE, main = "")
title(main = "Histogram of theta.1 and pdf of Gamma(12,6)")
# Then define a grid of values and add the line of the pdf values to plot
x.grid = seq(0.0001, 5, length = 1000)
lines(x.grid, dgamma(x.grid, shape = 12, rate = 6), col = "red")
```

(c) In the same way as for (b), plot a histogram of $\hat{\theta}_2$. Use the graphs to comment on the bias of the two estimators (you may also need to calculate the mean of each sampling distribution), and also on their relative efficiency (the variance of each sampling distribution will help). Use the following to obtain the two histograms side by side.

```
par(mfrow = c(1, 2))
hist(theta.1, xlim = c(0, 5), freq = FALSE)
hist(theta.2, xlim = c(0, 5), freq = FALSE)
```

3. Let $X_1, X_2, \ldots, X_n$ be a random sample from a $\mathrm{U}(0, \theta)$ distribution. We have shown in lectures that two estimators for θ are $\hat{\theta}_1 = 2\bar{X}$ and $\hat{\theta}_2 = X_{(n)}$. Let us generate many samples of size 6 with $\theta = 2$ and calculate many values of $\hat{\theta}_1$ and $\hat{\theta}_2$.

```
nsim=10^4
# Initialise vectors as arrays of 0's
theta.a=rep(0,nsim)
theta.b=rep(0,nsim)
for (i in 1:nsim){
x=runif(n=6,min=0,max=2)
theta.a[i] = 2 * mean(x)
theta.b[i] = max(x)
}
```

Calculate the mean of `theta.a` and `theta.b` and comment on bias. Do the same for variance and discuss the MSE of each estimator.

4. In lectures we showed if we have a random sample from a Gamma (α, β) distribution that suitable estimators for α and β are given by:

$$\hat{\alpha} = \frac{n\bar{X}^2}{(n-1)S^2}, \quad \hat{\beta} = \frac{n\bar{X}}{(n-1)S^2}$$

Calculate estimates for the values of α and β for the following sample:

3.9 3.4 5.2 2.6 3.4 2.7 3.1 2.4 4.2 1.7 2.4 3.3 2.3 2.0 3.4 2.9 2.5 3.9 2.3 1.1